

LOW ACID FOODS



Teens (14–18)

Q

Why must low acid foods like vegetables and meats be processed in a pressure canner rather than a boiling water canner?

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ANSWER

To reach temperatures higher than boiling water for killing harmful bacteria

FUN FACT

Pressure canners can heat your food above 240°F, which is needed to safely kill bacteria in low acid foods.

Source: EM 9435 Preserving dried beans

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Q

What is the typical pressure needed in a weighted gauge pressure canner to safely process low acid foods at sea level?

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ANSWER

**10 pounds per square inch
(psi)**

FUN FACT

The right pressure is important to reach the temperature needed for safe preservation!

Source: EM 9435 Preserving dried beans

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Teens (14–18)

Q

When using a dial gauge pressure canner for low acid foods, what pressure setting is recommended at sea level?

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ANSWER

**11 pounds per square inch
(psi)**

FUN FACT

Dial gauges need to be checked yearly to stay accurate for safe canning.

Source: EM 9435 Preserving dried beans

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Teens (14–18)

Q

How should pressure be adjusted in a pressure canner when canning low acid foods at altitudes between 2,001 and 4,000 feet?

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ANSWER

Increase pressure to 12 pounds (dial gauge) or 15 pounds (weighted gauge)

FUN FACT

Higher altitude means lower boiling temperature, so we increase pressure to keep food safe!

Source: EM 9435 Preserving dried beans

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What is the correct way to vent a pressure canner before sealing it to start timing?

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ANSWER

Let steam vent steadily for 10 minutes before placing weighted gauge or sealing lid

FUN FACT

Venting gets rid of air pockets so temperature can reach the safe level inside the canner.

Source: EM 9435 Preserving dried beans

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Why do jars of low acid foods need to be fully covered with water in the pressure canner?

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ANSWER

To ensure even heating and maintain correct temperature throughout processing

FUN FACT

Water transfers heat evenly so even the middle of the jar gets hot enough to be safe.

Source: EM 9435 Preserving dried beans



Q

How long should pint jars of dried beans be processed in a dial gauge pressure canner at 11 pounds pressure?

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ANSWER

65 minutes

FUN FACT

Low acid foods usually need long processing times to be safe at home.

Source: EM 9435 Preserving dried beans



Q

When processing jars of different sizes (pint and quart) together in a pressure canner, how do you determine the processing time?

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ANSWER

Use the time for the larger jar size

FUN FACT

Using the longer time ensures even the bigger jars are safe to eat later.

Source: PNW 450 Canning smoked fish at home

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Q

What should you do after removing the pressure canner from heat and before opening the lid when processing low acid foods?

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ANSWER

Wait until the pressure gauge drops to zero and then wait 10 minutes before opening

FUN FACT

Waiting prevents hot steam burns and lets jars finish firming safely.

Source: EM 9435 Preserving dried beans

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Q

Why are low acid foods at risk of bacteria growth if improperly canned?

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ANSWER

Because their pH is above 4.6, which does not stop dangerous spores from growing

FUN FACT

Low acid means bacteria can survive unless heat kills them in canning.

Source: EM 9435 Preserving dried beans

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If your pressure canner is more than 100 years old and has a dial gauge, what should you do before using it?

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ANSWER

Have the gauge tested for accuracy at your local Extension office or equipment tester

FUN FACT

An inaccurate pressure gauge can lead to unsafe food preservation.

Source: EM 9435 Preserving dried beans



Q

Why should you never open the pressure canner lid while it is still pressurized during processing?

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ANSWER

Opening releases steam rapidly and can cause serious burns

FUN FACT

Always wait for pressure to drop to zero before opening your canner!

Source: EM 9435 Preserving dried beans

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Q

What does 'headspace' mean when packing jars with low acid foods for canning?

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ANSWER

The space left between the top of the food and the jar rim

FUN FACT

Correct headspace helps jars seal properly and prevent spoilage.

Source: EM 9435 Preserving dried beans

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Teens (14–18)

Q

What must you NOT do when canning low acid foods to keep them safe?

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ANSWER

Use untested recipes or skip pressure canning steps

FUN FACT

Food safety depends on following tested methods exactly!

Source: EM 9435 Preserving dried beans

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Teens (14–18)

Q

How does altitude affect pressure canning of low acid foods?

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ANSWER

Higher altitude means increasing pressure to reach proper temperature

FUN FACT

Because water boils at lower temps higher up the mountain, we turn up the pressure to stay safe!

Source: EM 9435 Preserving dried beans

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Q

Which gauge type requires you to watch for specific weighted numbers to maintain pressure during canning?

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ANSWER

Weighted gauge pressure canner

FUN FACT

Weighted gauges jiggle or rock to keep the right pounds per square inch pressure.

Source: EM 9435 Preserving dried beans

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Teens (14–18)

Q

If using a weighted gauge canner above 1,000 feet altitude, what is the usual pressure setting for low acid food processing?

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ANSWER

15 pounds pressure

FUN FACT

*Weighted gauge settings are simple:
more weight equals more pressure.*

Source: EM 9435 Preserving dried beans

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Q

Why should jars not be stacked on top of each other when pressure canning low acid foods?

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ANSWER

Stacking jars can cause them to shift and break during processing

FUN FACT

Broken jars inside a canner are dangerous and waste food!

Source: PNW 450 Canning smoked fish at home

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Q

What is a safe headspace for canning dried beans with meat in quart jars?

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ANSWER

1 inch

FUN FACT

Correct headspace helps steam circulate during canning for safety.

Source: EM 9435 Preserving dried beans

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Teens (14–18)

Q

What temperature does a pressure canner reach at 10 pounds pressure that boiling water canners cannot?

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ANSWER

About 240°F (116°C)

FUN FACT

Temperatures above boiling water are necessary to kill bacterial spores in low acid foods.

Source: EM 9435 Preserving dried beans